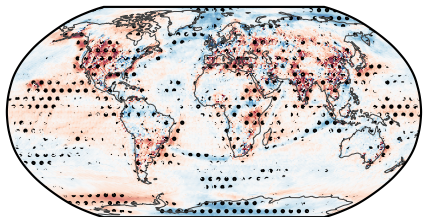


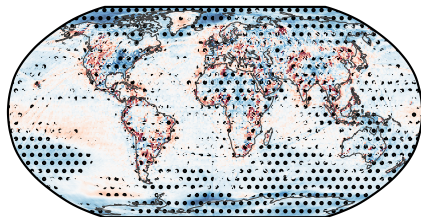
ENSEMBLE-MEAN emulator minus held-out CESM2 (5 emulator vs 6-11 CESM2 members), 10-year mean anomaly vs 1850-1900
 hatching = difference significant at $p < 0.05$ (Welch t-test, FDR-controlled at $q = 0.1$); precipitation as % of its 1850-1900 baseline

Historical
2005-2014 (10 yr)



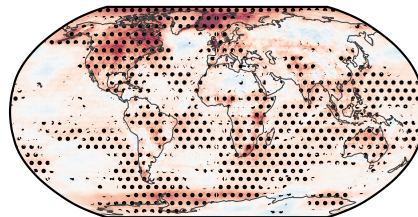
$r = 0.830$ RMSE = $0.401\text{ }^{\circ}\text{C}$
 30% of area significantly different

SSP3-7.0
2091-2100 (10 yr)



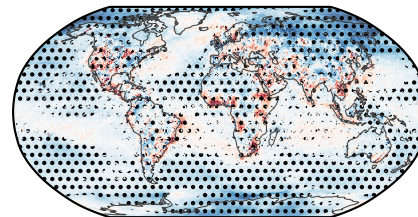
$r = 0.975$ RMSE = $0.425\text{ }^{\circ}\text{C}$
 44% of area significantly different

Aerosol-only (AAER)
2041-2050 (10 yr)

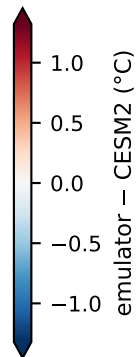


$r = 0.823$ RMSE = $0.338\text{ }^{\circ}\text{C}$
 47% of area significantly different

Greenhouse-gas-only (GHG)
2041-2050 (10 yr)

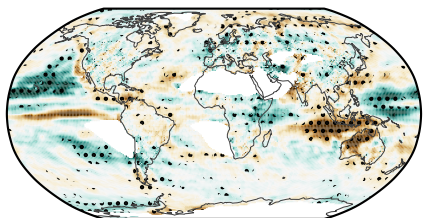


$r = 0.935$ RMSE = $0.414\text{ }^{\circ}\text{C}$
 54% of area significantly different

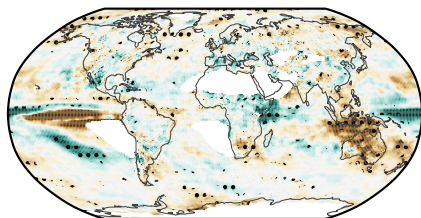


Temperature

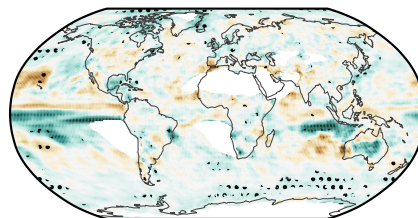
Precipitation



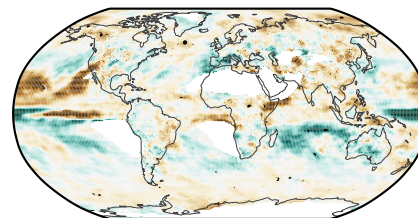
$r = 0.651$ RMSE = 6.710%
 18% of area significantly different
 (9% arid, masked)



$r = 0.981$ RMSE = 7.398%
 11% of area significantly different
 (9% arid, masked)



$r = 0.875$ RMSE = 4.724%
 5% of area significantly different
 (9% arid, masked)



$r = 0.962$ RMSE = 6.209%
 1% of area significantly different
 (9% arid, masked)

